

Challenge (Habanero)

Q1. Factor the trinomial $x^2 + 5x + 6$.

- (A) $(x + 2)(x + 3)$
- (B) $(x + 1)(x + 6)$
- (C) $(x + 3)(x + 2)$
- (D) $(x - 2)(x - 3)$
- (E) $(x - 1)(x - 6)$

Q2. A construction company is building a rectangular garden with an area of $x^2 + 7x + 12$ square meters. What are the factors of the area?

- (A) $(x + 3)(x + 4)$
- (B) $(x + 2)(x + 6)$
- (C) $(x + 1)(x + 12)$
- (D) $(x - 3)(x - 4)$
- (E) $(x - 2)(x - 6)$

Q3. Factor the trinomial $x^2 - 4x - 5$.

- (A) $(x - 5)(x + 1)$
- (B) $(x + 5)(x - 1)$
- (C) $(x + 1)(x - 5)$
- (D) $(x - 1)(x + 5)$
- (E) $(x + 4)(x - 4)$

Q4. A recipe for making cookies requires $x^2 + 2x + 1$ cups of flour. What are the factors of the amount of flour?

- (A) $(x + 1)^2$
- (B) $(x - 1)^2$
- (C) $(x + 2)(x + 1)$
- (D) $(x - 2)(x - 1)$
- (E) $(x + 1)(x - 1)$

Q5. Factor the trinomial $x^2 - 9x + 20$.

- (A) $(x - 5)(x - 4)$
- (B) $(x + 5)(x + 4)$
- (C) $(x - 4)(x - 5)$
- (D) $(x + 4)(x - 5)$
- (E) $(x - 1)(x - 20)$

Q6. A builder is constructing a wall with an area of $x^2 + 10x + 24$ square meters. What are the factors of the area?

- (A) $(x + 4)(x + 6)$
- (B) $(x + 2)(x + 12)$
- (C) $(x + 1)(x + 24)$
- (D) $(x - 4)(x - 6)$
- (E) $(x - 2)(x - 12)$

Q7. Factor the trinomial $x^2 + x - 6$.

- (A) $(x + 3)(x - 2)$
- (B) $(x + 2)(x - 3)$
- (C) $(x + 1)(x - 6)$
- (D) $(x - 1)(x + 6)$
- (E) $(x - 2)(x + 3)$

Q8. A chef is preparing a dish that requires $x^2 - 3x - 4$ cups of ingredients. What are the factors of the amount of ingredients?

- (A) $(x - 4)(x + 1)$
- (B) $(x + 4)(x - 1)$
- (C) $(x + 1)(x - 4)$
- (D) $(x - 1)(x + 4)$
- (E) $(x - 2)(x + 2)$

Open Ended Questions (FRQ)

Q9. Factor the trinomial $x^2 + 8x + 15$ and explain the steps you took to factor it.

Q10. A contractor is building a rectangular foundation with an area of $x^2 + 9x + 20$ square meters. Factor the area and determine the dimensions of the foundation.

Chef's Tips

- Read the questions carefully
- Use the correct factoring method
- Check your work